Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Reads graph and uses $10^{\log P}$ or $10^{\log V}$ to get either P or V	AO3.4	M1	$\log_{10}P = 2.18$ $P = 151$ $\log_{10}V = -0.15$ $V = 0.708$
	Correctly obtains P and V AWRT 150 and AWRT 0.71	AO1.1b	A1	
(b)	Calculates value of gradient to find d Condone use of log d = gradient Or uses a value of c plus a P/V pair to find d	AO3.4	M1	$\log_{10}P = \log_{10} c + d \log_{10}V$
	Obtains correct value for d AWRT −1.4 Not necessarily a decimal	AO1.1b	A1	Gradient = $d = -1.4$
	Calculates value of intercept to find $\log_{10} c$ Or uses a value of d plus a P/V pair to find c	AO3.4	M1	Intercept = $\log_{10} c$
	Calculates correct value for c AWRT 93	AO1.1b	A1	$c = 93.3$
(c)	Uses their values of c and d in the formula $P = cV^d$	AO1.1a	M1	$P = 93.3 \times 2^{-1.4}$
	Obtains P value, **including units** AWFW 30 to 40	AO3.2a	A1	= 35.4 kilopascals
				Total 8 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains (at least four) correct $\log_{10}y$ values, in	AO1.1a	M1	(1, 1.1) (2, 1.7) (3, 2.1) (4, 3.0)

	table or plotted			(5, 3.1) (6, 3.5)
	Plots **all** points correctly	AO1.1b	A1	(Points above plotted on grid)
(b)	Identifies y = 1100 and gives correct reason	AO2.2b	B1	(4, 1100), as it is not on the line that the other points are close to
(c)	Uses laws of logs. (May earn in part **(a)** if laws of logs were used there)	AO1.1a	M1	$\log_{10}y = \log_{10}k + x\log_{10}b$ Vertical intercept c = 0.68 ($=\log_{10}k$) Therefore from intercept: $k = 10^{0.68}$
	Draws straight line **and** calculates/measures the vertical intercept c and attempts 10^c **or** calculates/measures gradient m and attempts 10^m Alternatively uses regression line from calculator to get intercept and gradient	AO1.1a	M1	Gradient m = 0.48 $=\log_{10}b$ Therefore from gradient: $b = 10^{0.48}$
	Finds correct value of b from 'their' gradient, provided 0.45 < 'their' gradient < 0.51	AO1.1b	A1F	k = 4.8
	Finds correct value of k from 'their' intercept, provided 0.6 ≤ 'their' intercept ≤ 0.8	AO1.1b	A1F	b = 3.0
				Total 7 marks

Q3.

Marking Instructions	AO	Marks	Typical Solution
Correctly applies a single law of logs with either term	AO1.1a	M1	$\log_a(a^3) + \log_a\left(\frac{1}{a}\right) = 3 + (-1)$ $= 3 - 1$
States correct final answer (NMS scores full marks)	AO1.1b	A1	

			= 2
			Total 2 marks

Q4.

Marking Instructions	**AO**	**Marks**	**Typical Solution**
States the correct gradient of the curve	AO1.2	B1	Grad of curve = $2e^{2x}$
Forms an equation using 'their' gradient of the curve and puts it equal to $\frac{1}{2}$	AO1.1a	M1	= grad of tangent so $2e^{2x} = \frac{1}{2}$
Takes a log of each side of 'their' equation and uses law of logs to obtain equation in x	AO1.1a	M1	$e^{2x} = \frac{1}{4} \Rightarrow 2x = \ln\left(\frac{1}{4}\right)$
Obtains a correct exact value for x	AO1.1b	A1	$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1}{4}\right) = \ln\left(\frac{1}{2}\right) = -\ln 2$
Substitutes 'their' value of x and obtains y value and hence the coordinates (follow through provided values are exact)	AO1.1b	A1F	$y = e^{2x} = \frac{1}{4}$ $\left(-\ln 2, \frac{1}{4}\right)$
			Total 5 marks

Q5.

	Marking Instructions	**AO**	**Marks**	**Typical Solution**
(a)(i)	States correct value CAO	AO3.4	B1	50
(ii)	States correct integer value CAO	AO3.4	B1	609
(b)	Forms correct equation and rearranges to obtain $e^{0.5t} = \ldots$	AO3.4	M1	$150 = 50e^{0.5t}$ so $e^{0.5t} = 3$
	Obtains the correct solution. Must give answer to 3 sf	AO1.1b	A1	$t = 2\ln 3 = 2.20$

(c)	1 mark for any clear valid reason, must be set in context of the question	AO3.5b	E1	No constraint on the number of rabbits (ie could go off to infinity) **OR** Model is only based on the 3 years of the study. Things may change **OR** Continuous model but number of rabbits is discrete **OR** Ignores extraneous factors such as disease, predation, limited food supply
(d)	Forms an equation with exponentials by letting $R = C$ PI	AO3.4	M1	$1000e^{0.1t} = 50e^{0.5t}$ $20 = e^{0.4t}$
	Solves 'their' equation correctly	AO1.1a	M1	$t = \ln 20 \div 0.4$ $= 7.49$
	States correct answer as the year 2023 CAO NMS scores full marks for 2023	AO3.2a	A1	2023
				Total 8 marks

Q6.

$\log_a N - \log_a x = \frac{3}{2}$

$\log_a \frac{N}{x} = \frac{3}{2}$

A log law used correctly. PI by next line.

M1

$\frac{N}{x} = a^{\frac{3}{2}}$

Logarithm(s) eliminated correctly

m1

$x = a^{-\frac{3}{2}} N$

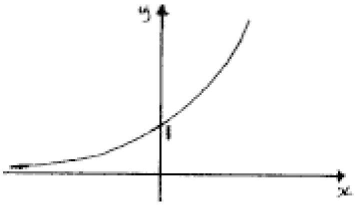
ACF of RHS

A1

[3]

Q7.

(a)



Correct graph, must clearly go below the intersection pt and an indication of correct behaviour of curve for large positive and large negative values of x.

Ignore any scaling on axes.

B1

Only one y−intercept, marked / stated as 1 or as coords (0, 1) with graph having no other intercepts on either axes.

B1

2

(b) $9^x = 15 \Rightarrow x \log 9 = \log 15$

OE eg $x = \log_9 15$

M1

(x =) 1.23(2486…) = 1.23 to 3sf

Condone > 3sf.

Must see evidence of logs used so NMS scores 0/2

A1

2

(c) {f(x) =} 9^{-x}

OE

B1

1

[5]

Q8.

(a) $b = a^c$

B1

1

(b) $2 \log_2 (x + 7) - \log_2 (x + 5) = 3$

$\log_2 (x + 7)^2 - \log_2 (x + 5) = 3$

A law of logs used correctly on a correct expression.

M1

$\log_2 \dfrac{(x+7)^2}{x+5} = 3$

A further correct use of law of logs on a correct expression.

M1

$= 3 \log_2 2 = \log_2 2^3$

$\Rightarrow \dfrac{(x+7)^2}{x+5} = 2^3$

$3 = 3 \log_2 2$ or $3 = \log_2 2^3$ ($= \log_2 8$) seen
or
eg $\log f(x) = 3 \Rightarrow f(x) = 2^3$ ($= 8$) OE

B1

$\Rightarrow (x + 7)^2 = 8(x + 5)$

Correct equation having eliminated logs and fractions

A1

$\Rightarrow x^2 + 14x + 49 = 8x + 40$
$\Rightarrow x^2 + 6x + 9$ $(= 0)$

A1

Since $6^2 - 4(1)(9) = 0$, (there is only) one value of x (which satisfies the given equation).

OE
CSO Need conclusion which is also correctly justified

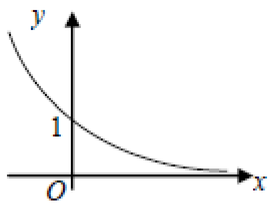
A1

6

[7]

Q9.

(a)



Correct shaped graph in 1st two quadrants only and indication of correct behaviour of curve for large positive and negative vals. of x. Ignore any scaling on axes.

B1

y-intercept indicated as 1 on diagram or stated as intercept = 1 or as coords (0, 1).

B1

2

(b) $\frac{1}{2^x} = \frac{5}{4} \Rightarrow 2^{-x} = \frac{5}{4}$ (or $2^x = \frac{4}{5}$ or $2^{2-x} = 5$)

Correct 'rearrangement' to eg $2^x = \frac{4}{5}$ or $2^{-x} = \frac{5}{4}$ or $0.5^x = 1.25$ PI or $\log 1 - \log 2^x = \log(5/4)$ or better

M1

$\log 2^{-x} = \log 1.25 \Rightarrow -x\log 2 = \log 1.25$

[$\log 2^x = \log 0.8 \Rightarrow x \log 2 = \log 0.8$]

[$\log 2^{2-x} = \log 5 \Rightarrow (2 - x) \log 2 = \log 5$]

[$2^x = 0.8$, $x\log_2 0.8$]; [$0.5^x = 1.25$, $x = \log_{0.5} 1.25$]

Takes logs of both sides of eqn of form either $2^x = k$ or $2^{-x} = k$ OE and uses 3rd law of logs or log to base 2 (or base ½) correctly

M1

$x = -0.321928$ … so $x = -0.322$ (to 3sf)

*Condone > 3sf [Logs must be seen to be used otherwise **max** of M1M0A0]*

A1

3

(c) $\log_a b^2 + 3\log_a y = 3 + 2\log_a \left(\frac{y}{a}\right)$

$\log_a b^2 + 3\log_a y = 3 + 2 [\log_a y - \log_a a]$

A log law used correctly; condone missing base a.

M1

$\log_a b^2 + 3\log_a y = 3 - 2\log_a a$

A different log law used correctly condone missing base a.

M1

$\log_a b^2 y = 3 - 2(1)$ [or $\log_a b^2 y + \log_a a^2 = 3$]

Either a further different log law used correctly condone *missing base a or $\log_a a = 1$ a stated / used.*

M1

$\Rightarrow \log_a b^2 y = 1 \quad \Rightarrow \; b^2 y = a$

$\log Z = k \Rightarrow Z = a^k$ used or a correct method to eliminate logs (dep on no misapplication of any log law OE in the whole solution) Rearrangements which require only two of the above Ms toeliminate logs correctly: award the remaining M with the m mark.

M1

$\Rightarrow \quad y = ab^{-2}$

ACF of RHS

A1

5

[10]

Q10.

(a)

stretch I
SF 4 II
in y-direction III

translate $\begin{pmatrix} e \\ 0 \end{pmatrix}$

either order

$I + (II \text{ or } III)$

M1A1

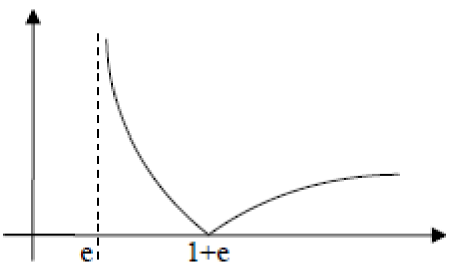
E1

accept 'e in positive x-direction'

B1

4

(b)



mod graph, in 2 connected sections, both in the first quadrant, touching x-axis

M1

curve touches x-axis at 1 + e (or 3.7 or better), and labelled (ignore scale)

A1

correct curvature, including at their 1 + e, approx. asymptote at $x = e$

A1

3

(c) (i) $|\,4\ln(x - e)\,| = 4$

$4\ln(x - e) = 4$ $\quad$ $4\ln(x - e) = -4$ or better

must see 2 equations, condone omission of brackets

M1

$(x =)\ 2e$ do not ISW

accept values of AWRT 5.42, 5.43, 5.44

A1

$(x =)\ e + e^{-1}$ or $(x =)\ e + \frac{1}{e}$ do not ISW

accept values of AWRT 3.08, 3.09

if M0 then x = 2e with or without working scores SC1

A1

3

(ii) $x \geq 2e$

accept values of AWRT 5.42, 5.43, 5.44

B1

$e < x \leq e + \frac{1}{e}$

accept values of AWRT 2.72, 3.08, 3.09

if B2 not earned, then SC1for any of

$e \leq x \leq e + \frac{1}{e}$, $e < x < e + \frac{1}{e}$, $e \leq x < e + \frac{1}{e}$

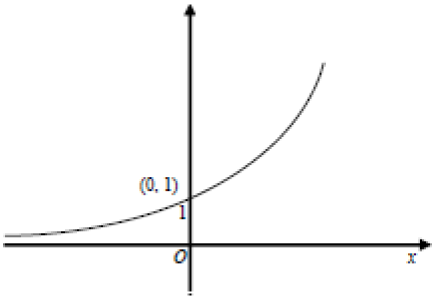
B2

3

[13]

Q11.

(a)



Correct shape, curve in 1st two quadrants only, crossing positive y-axis once and asymptotic to negative x-axis.

B1

Coordinates (0, 1). Accept y-intercept indicated as 1 on diagram or stated as 'intercept = 1' B0 if graph clearly drawn crossing axes at more than one point

B1

2

(b) (i) $y^2 - 12 = y$ OE; $7^{2x} - 12 = 7^x$ OE

Eliminates either x or y correctly

M1

$(y - 4)(y + 3)$ (= 0); $(7^x - 4)(7^x + 3)$ (= 0)

Correct factors or $y = \dfrac{1 \pm \sqrt{49}}{2}$ or better or $7^x = \dfrac{1 \pm \sqrt{49}}{2}$ or better

A1

Since y (= 7^x) > 0, [y (=7^x) ≠ − 3]
(there is exactly one point of intersection)

Clear indication that c's negative solution(s) has / have been considered and rejected

E1

y-coordinate is 4

B1

4

(ii) $7^x = 4$ so $x \log 7 = \log 4$ [or $x = \log_7 4$]

OE ft on $7^x = k$, where k is positive, to either $x \log 7 = \log k$ or $x = \log_7 k$

M1

x = 0.712(414…) = 0.712 to 3SF

Condone > three significant figures. If use of logarithms not explicitly seen then score 0 / 2

A1

2

[8]

Q12.

(a) (i) (p =) 3

B1

1

(ii) (q =) –3

If not correct, ft on −p

B1F

1

(iii) $(r =)\frac{1}{2}$

OE

B1

1

(b) $2^{\frac{1}{2}} \times 2^{x} = 2^{-3} \Rightarrow 2^{\frac{1}{2}+x} = 2^{-3}$

Using a law of indices or logs correctly to combine at least two of the powers of 2 PI

M1

$\Rightarrow x = -3\frac{1}{2}$

If not correct, ft on $x = q - r$ provided method shown

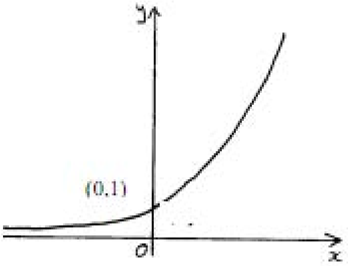
A1F

2

[5]

Q13.

(a)



*Any graph only crossing the y–axis at (0, 1) stated /indicated … (accept 1 on y–axis as equivalent) and not drawn **below** x–axis*

B1

Correct shaped graph, must clearly go below the intersection pt and an indication of correct behaviour of curve for large positive and large negative values of x. Ignore any scaling on axes.

B1

2

(b) Translation;

Accept 'transl…' as equivalent
[T or Tr is NOT sufficient]

B1

$$\begin{bmatrix} 0 \\ -5 \end{bmatrix}$$

*If vector not given, accept **full** equivalent to vector in words provided linked to 'transl../move/shift'*
(B0B0 if >1 transformation)

B1

2

(c) (i) $4^x = (2^2)^x = 2^{2x} = (2^x)^2 = Y^2$

$2^{x+2} = 2^x \times 2^2 = 4Y$

Justifying either $4^x = Y^2$ or $2^{x+2} = 4Y$ with no errors seen

M1

$$4^x - 2^{x+2} - 5 = 0 \Rightarrow Y^2 - 4Y - 5 = 0$$

AG Be convinced; must have justified both of the above.

A1

2

(ii) $(Y - 5)(Y + 1) = 0$

Correct factorising or use of quadratic formula or completing sq. PI by both solns 5 & −1 seen

M1

(Since) $2^x > 0$ (for all real x,) $2^x = 5$ so only one (real) solution

Rejection of 2^x (condone Y) negative, with justification, (condone "2^x not negative") followed by statement

E1

$\log 2^x = \log 5 \Rightarrow x \log 2 = \log 5$

Eqn of form $p^x = q \Rightarrow x\log p = \log q$ provided $p > 0$ & $q > 0$ OE eg $x = \log_2 5$

M1

$x = 2.3219\ldots = 2.322$ (to 3dp)

Condone > 3dp but must see explicit use of logs and must only be the one solution.

A1

4

[10]

Q14.

(a) $3 \ln x = 4$

$\left(\ln x = \frac{4}{3}\right)$

$x = e^{\frac{4}{3}}$

ISW. Condone $\sqrt[3]{e^4}$

B1

1

(b) $3\ln x + \frac{20}{\ln x} = 19$

$3(\ln x)^2 + 20 = 19 \ln x$

correctly multiplying by ln x.

M1

$3(\ln x)^2 \ -19 \ln x + 20 \ (= 0)$

A1

$(3\ln x \pm 4)(\ln x \pm 5) \ (= 0)$

use of formula, or completing the square must be correct

m1

$\ln x = \frac{4}{3}, 5$

A1

$x = e^{\frac{4}{3}}, e^5$

condone $\sqrt[3]{e^4}$

A1

5

[6]

Q15.

(a) (i) $(x =)$ 1

CAO

B1

1

(ii) $(x =)$ 3

CAO

B1

1

(b) $\log_a n^2 = \log_a 18(n - 4)$

A valid law of logs applied to correct logs

M1

A second valid law of logs applied to correct logs

M1

$n^2 - 18n + 72 = 0$

ACF of these terms eg $n^2 - 18n = -72$

A1

$(n - 6)(n - 12) = 0$

Valid method to solve quadratic, dep on both the previous Ms

m1

$n = 6$, $n = 12$

Both values required
***SC** NMS max (out of 5) B3 for both 6 and 12 without uniqueness considered; max B1 for either 6 or 12 only*

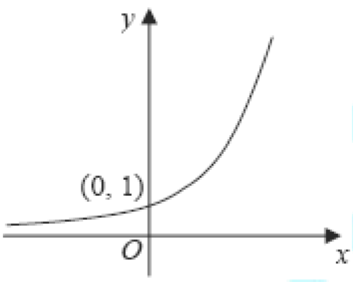
A1

5

[7]

Q16.

(a)



Shape with some indication of asymptotic behaviour in 2nd quadrant below pt of intersection with y-axis

B1

Only intersection is with y-axis at (0, 1) stated/indicated … (accept 1 on y-axis as equivalent)

B1

2

(b) (i) $h = 0.5$

PI

B1

$f(x) = 2^x$
$I \approx h/2\{\ldots\}$
$\{\ldots\} = f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]$

OE summing of areas of the 4 'trapezia'

M1

$\{\ldots\} = 1 + 4 + 2(\sqrt{2} + 2 + \sqrt{8})$
$= 5 + 2 \times 6.2426\ldots = 17.485\ldots$

OE Accept 2 dp (rounded or truncated) as evidence for surds

A1

$(I \approx)\ 4.3713\ldots = 4.37$ (to 3 sf)

CAO Must be 4.37
***SC** for those who use 5 strips, max possible is B0M1A1A0*

A1

4

(ii) Increase the number of ordinates

OE

E1

1

(c) Translation;

Accept 'translat…' as equivalent
[T or Tr is NOT sufficient]

B1;

$\begin{bmatrix} -7 \\ 3 \end{bmatrix}$

B1 for each component of the vector.
Condone if the equiv 2 vectors are given.
*Accept **full** equivalent to vector(s) in words provided linked to 'translation/ move/shift' and **correct** directions.*
*(No marks if **different** transformations)*

B1;B1

3

(d) $8 = 2^k + 3 \Rightarrow 2^k = 5$

Correct subst. and an attempted rearrangement to $2^k = N$. PI by $k = \frac{\log 5}{\log 2}$

M1

$k = \log_2 5$

Accept $m = 2$, $n = 5$

A1

2

[12]

Q17.

(a) (i) $t = 0$ $h = A(1 - 1) = 0$

B1

1

(ii) $57 = A\left(1 - e^{\frac{12}{4}}\right)$

M1

$A = \dfrac{57}{(1 - e^{-1})} \approx 60$

Or 59.9… seen.

A = correct expression ≈ 60 2 sf

A1

2

(b) (i) $h = 48$ $\dfrac{48}{60} = 1 - e^{-\frac{1}{4}t}$

M1

$\ln\left(e^{-\frac{1}{4}t}\right) = \ln\left(\dfrac{1}{5}\right)$

m1

$-\dfrac{1}{4}t = -\ln 5 \Rightarrow t = 4\ln 5$

A1

3

(ii) $\dfrac{dh}{dt} = -\dfrac{1}{4} \times -60 \times e^{-\frac{1}{4}t}$

Differentiate, condone sign errors

M1

$60e^{-\frac{1}{4}t} = 60 - h \Rightarrow \dfrac{dh}{dt} = \dfrac{1}{4}(60 - h)$

Eliminate $e^{-\frac{1}{4}t}$

m1

$\frac{dh}{dt} = 15 - \frac{h}{4}$

CSO, AG

A1

3

(iii) $h = 8$

B1

1

[10]

Q18.

(a) (i) $y = \ln(5x - 2)$

$\frac{k}{5x-2}$

M1

$\left(\frac{dy}{dx} =\right)\frac{5}{5x-2}$

No ISW, eg $\frac{5}{5x-2} = \frac{1}{x-2}$ *(M1A0)*

A1

2

(ii) $y = \sin 2x$

$\left(\frac{dy}{dx} =\right)2\cos 2x$

A1

2

$k \cos 2x$

M1

(b) (i) $f(x) \geq \ln 0.5$ or $f(x) \geq -\ln 2$

M1A1

2

(ii) $(gf(x) =)$ $\sin [2 \ln (5x - 2)]$

or (gf(x) =) sin ln $(5x - 2)^2$

Condone
sin 2 ln (5x – 2) or sin 2 (ln (5x – 2))
*but **not** sin 2 (ln 5x – 2) or sin 2 ln 5x – 2*

B1

1

(iii) gf(x) = 0

sin[2 ln($5x - 2$)] = 0

2 ln($5x - 2$) = 0

Correct first step from their (b)(ii)

M1

$5x - 2 = 1$

Their f(x) = 1 from k ln (f(x)) = 0

m1

$x = \frac{3}{5}$

Withhold if clear error seen other than omission of brackets

A1

3

(iv) $x = \sin 2y$

$\sin^{-1} x = 2y$ (or $\sin^{-1} y = 2x$)

Correct equation involving sin^{-1}

M1

($g^{-1}(x)$ =) $\frac{1}{2}\sin^{-1} x$

A1

2

[12]

Q19.

(a) (i) $\log_a 40$

Accept 'k = 40'

B1

1

(ii) $\log_a 8$

Accept 'k = 8'

B1

1

(iii) $\log_a 125$

Accept 'k = 125' but not '$k = 5^3$'

B1

1

(b) $\log_{10} \{(1.5)^{3x}] = \log_{10} 7.5$

Correct statement having taken logs of both sides of $(1.5)^{3x} = 7.5$ OE PI or $3x = \log_{1.5} 7.5$ seen

M1

$3x \log_{10} 1.5 = \log_{10} 7.5$

$\log 1.5^{3x} = 3x \log 1.5$ OE

m1

$x = \dfrac{\lg 7.5}{3 \lg 1.5} = 1.65645\ldots = 1.656$ to 3 dp

Both method marks must have been awarded with clear use of logarithms seen

A1

3

(c) $\log_2 p = m \Rightarrow p = 2^m$; $\log_8 q = n \Rightarrow q = 8^n$

Either $p = 2^m$ or $q = 8^n$ seen or used

M1

$p = 2^m$ and $q = 2^{3n}$

Writing $8^n = 2^{3n}$ and having $p = 2^m$

m1

$pq = 2^m \times (2^3)^n = 2^m \times 2^{3n}$ so $pq = 2^{m+3n}$

Accept $y = m + 3n$

A1

3

[9]

Q20.

(a) $3e^x = 4$

$e^x = \frac{4}{3}$

M1

$x = \ln \frac{4}{3}$

A1

2

(b) (i) $3e^x + 20e^{-x} = 19$

$3y + \frac{20}{y} = 19$ or $3e^{2x} + 20 = 19e^x$

$3y^2 - 19y + 20 = 0$

AG

B1

1

(ii) $(3y - 4)(y - 5) = 0$

$y = \frac{4}{3}, 5$

B1

$\therefore x = \ln\frac{4}{3}, \ln 5$

ln (their + ve y's)

M1A1

3

[6]

Q21.

(a) (i) $\sqrt{125} = \sqrt{25 \times 5} = 5\sqrt{5}$

OE eg $\sqrt{125} = \sqrt{5^3}$ *or* $5^{1.5}$ *seen*

M1

$5^p = \sqrt{125} \Rightarrow p = 1.5$

Correct value of p must be explicitly stated

A1

Alternative:

EXAM PAPERS PRACTICE

$$p\log 5 = \frac{1}{2}\log 125$$

2

OE eg p log 5 = log 11.18

or eg $p = \log_5 \sqrt{125}$

(M1)

$$p\log 5 = \frac{3}{2}\log 5 \Rightarrow p = \frac{3}{2}$$

Correct value of p must be explicitly stated

(A1)

(ii) $5^{2x} = \sqrt{125} = 5^p \Rightarrow x = 0.5p = 0.75$

Must be 0.5 × c's value of p

SC: x = 0.75 with c's ans (a)(i) $5^{1.5}$ scores B1F

B1F

1

(b) $3^{2x-1} = 0.05$

$(2x - 1)\log 3 = \log 0.05$

Take logs of both sides and use 3rd law of logs. PI eg by $2x - 1 = \log_3 0.05$ seen

M1

$$x = \frac{\log_{10} 0.05}{2\log_{10} 3} + \frac{1}{2}$$

Correct rearrangement to x = ... PI

m1

= –0.8634(165...) = –0.8634 to 4dp

*Condone > 4 dp. Must see logs clearly **used** in solution, so NMS scores 0/3*

A1

3

(c) $\log_a x = 2(\log_a 3 + \log_a 2) - 1$

$= 2\log_a (3 \times 2) - 1$

A valid law of logs used

M1

$= \log_a (6^2) - 1$

Another valid law of logs used

M1

$= \log_a 36 - \log_a a$

$\log_a a = 1$ quoted or used

or $\log_a \frac{x}{k} = -1 \Rightarrow \frac{x}{k} = a^{-1}$ *OE*

B1

$\log_a x = \log_a \left(\frac{36}{a}\right) \Rightarrow x = \frac{36}{a}$

CSO Must be $x = \frac{36}{a}$ or $x = 36a^{-1}$

A1

4

[10]

Q22.

(a) $2 \ln x = 5$

$\ln x = \frac{5}{2}$ $x = e^{\frac{5}{2}}$

B1

1

(b) $2 \ln x + \frac{15}{\ln x} = 11$

$2(\ln x)^2 - 11 \ln x + 15 = 0$

Forming quadratic equation in ln x
condone poor notation

M1

$(2 \ln x - 5)(\ln x - 3) = 0$

Attempt at factorisation/formula

m1

$\ln x = \frac{5}{2}, 3$ condone $2 \ln x = 5$

A1

$x = e^{\frac{5}{2}}, e^{3}$

[SC for substituting $x = e^{\frac{5}{2}}$ *or equivalent*

into equation and verifying B1 $\left(\frac{1}{5}\right)$ *]*

A1, A1

5

[6]

Q23.

(a) $x = 8$

No clear log law errors seen.

Condone answer left as $\frac{16}{2}$

B1

1

(b) $\log_a y = \log_a 3^2 + \log_a 4 + 1$

One law of logs used correctly

M1

$\log_a y = \log_a (3^2 \times 4) + 1$

Either a second law of logs used correctly
or the 1 written as $\log_a a$

M1

$\log_a y = \log_a (3^2 \times 4) + \log_a a = \log_a 36a$

$\Rightarrow y = 36a$

CSO

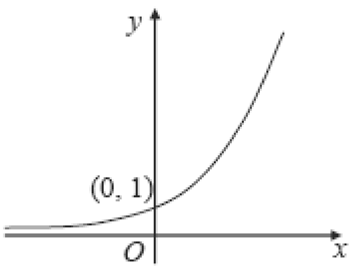
A1

3

[4]

Q24.

(a)



Shape (graph must clearly go below the intersection pt.). Condone if x-axis is a tangent

B1

Only intersection with y-axis at (0, 1) stated/ indicated … (accept 1 on y-axis as equivalent) 0

B1

2

(b) (i) Stretch **(I)** in x-direction **(II)** scale factor 0.5 **(III)**

*Need **(I)** & one of **(II), (III)***

M1

M0 if >1 transformation

A1

2

(ii) Translation;

Must be 'Translation' or 'translate (d)' for 1st B mark

B1;

$\begin{bmatrix} -1 \\ 0 \end{bmatrix}$

*Accept **full** equivalent to vector in words provided linked to 'translation/ move/shift' and **negative** x-direction (Note: B0 B1 is possible)*

B1

ALTn: Stretch **(I)** in y-direction **(II)** scale factor 3 **(III)**

[Mark the alternative as in (b)(i).]

2

(c) (i) $9^x = (3^2)^x = 3^{2x} = (3^x)^2 = Y^2$;

Justifying either $9^x = Y^2$ or $3^{x+1} = 3Y$

M1

$3^{x+1} = 3^x \times 3^1 = 3Y$

$9^x - 3^{x+1} + 2 = 0 \Rightarrow Y^2 - 3Y + 2 = 0$

$\Rightarrow (Y-1)(Y-2) = 0$

AG

A1

2

(ii) $Y = 1 \Rightarrow 3^x = 1 \Rightarrow x = 0$

AG (Accept direct substitution if convinced)

B1

$Y = 2 \Rightarrow 3^x = 2$

$\log_{10} 3^x = \log_{10} 2$

Takes logs of both, PI by 'correct' value(s) later.
or $x = \log_3 2$ seen

M1

$x \log_{10} 3 = \log_{10} 2$

Use of log $3^x = x$ log 3 or

$\log_3 2 = \frac{\lg 2}{\lg 3}$ OE (PI by $\log_3 2 = 0.630$ or 0.631 or better)

m1

$x = \frac{\lg 2}{\lg 3} = 0.630929 \ldots = 0.6309$ to 4 dp

Must show that logarithms have been used otherwise 0/3

A1

4

[12]

Q25.

(a) (i) $A = 20$

B1

1

(ii) $\frac{2000}{A} = k^{60}$

M1

$k = (100)^{\frac{1}{60}} = 1.079775$

AG, or $k = 10^{\frac{\log 100}{60}} = 10^{0.0333}$ or $\sqrt[60]{100}$ or $\sqrt[30]{10}$ or $e^{\frac{\ln 100}{60}} = e^{0.076}$ or $e^{0.077}$ or 1.0797751(6) seen

A1

2

(iii) $P = 20 \times k^{2008-1885}$

M1

$= 251780 \approx 252000$

CAO mearest 1000

A1

2

(b) $15 \times 1.082709^t = 20 \times 1.079775^t$

equate prices

M1

$$\frac{15}{20} = \left(\frac{1.079775}{1.0822709}\right)^t$$

t as a single index, or correct log expression at this stage

M1

$$t = \frac{\log 0.75}{\log 0.997290}$$

expression for t

m1

$t = 106.017 \Rightarrow 1991$

***SC** Answer only/Trial and error*
106 seen (2 out of 4)
1991 (4 out of 4)

A1

4

[9]

Q26.

(a) (i) $\log_a 1 = 0$

B1

1

(ii) $\log_a a = 1$

B1

1

(b) $\log_a x = \log_a (5 \times 6) - \log_a 1.5$

One law of logs used correctly

M1

$\log_a x = \log_a \left(\frac{5 \times 6}{1.5}\right)$

A second law of logs used correctly

M1

$\log_a x = \log_a 20 \Rightarrow x = 20$

A1

3

[5]

Q27.

(a) (i) $h = 0.5$

Integral = $h/2$ {……}

PI

B1

$\{..\} = f(0) + 2[f\left(\frac{1}{2}\right) + f(1) + f\left(\frac{3}{2}\right)] + f(2)$

OE summing of areas of the four traps.

M1

$\{\ \} = 1 + 2[\sqrt{6} + 6 + 6\sqrt{6}] + 36$

= 1+2 [2.449.. + 6 + 14.6969..] + 36

Condone 1 numerical slip. Accept 3sf values if not exact.

A1

= 37 + 2 × 23.146.. = 83.292…

Integral = 0.25 × 83.292.. = 20.8 (3sf)

CAO; must be 20.8

A1

4

(ii) Relevant trapezia drawn on a copy of given graph

Accept single trapezium with its sloping side above the curve

M1

{Approximation is an} overestimate

Dep. on 4 trapezia with each of their upper vertices lying on the curve

A1

2

(b) (i) Stretch **(I)** in x-direction **(II)**

*Need **(I)** and one of **(II)**, **(III)***
M0 if more than one transformation

M1

(scale factor) $\frac{1}{3}$ **(III)**

A1

2

(ii) $6^{3x} = 84$

PI

M1

$\log_{10} 6^{3x} = \log_{10} 84$

Take logs of both sides of $a^x = b$, PI by 'correct' value(s) later or $3x = \log_6 84$

M1

$3x \log_{10} 6 = \log_{10} 84$

Use of $3 \log 6^{3x} = 3x \log 6$ OE or $3x = \log_6 84$ seen

m1

$x = \frac{\lg 84}{3 \lg 6}$

$x = 0.82429.... = 0.824$ (to 3dp)

Must see that logs have been used before any of the last 3 marks are awarded in (b)(ii). Condone > 3dp

A1

4

(c) $f(x) = 6^{x-1} - 2$

B1For either $6^{x-1} + 2$ or for $6^{x+1} - 2$

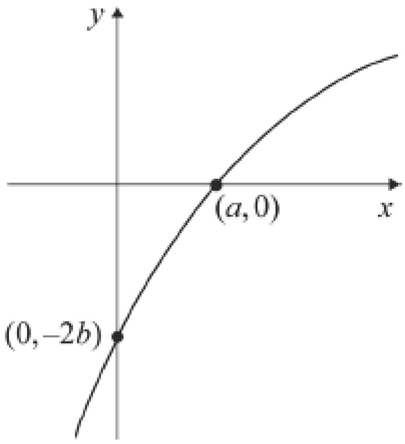
B2,1

2

[14]

Q28.

(a)



shape

B1

coordinates

B1

2

(b) (i) Translation

M1

$\begin{bmatrix} -1 \\ 0 \end{bmatrix}$

A1

Stretch I

I + (II or III)

M1

SF 4 II

// y-axis III

I + II + III

A1

Translation $\begin{bmatrix} 0 \\ -2 \end{bmatrix}$

both

B1

All correct and no mistakes on order etc

A1

OR

I stretch M1 I + (II or III)

II SF 4

III // y-axis A1 (I + II + III)

Translation M1

$\begin{bmatrix} -1 \\ -2 \end{bmatrix}$ *A1* *B1*

All correct A1

6

Alternative:

$$y = 4\ln(x+1) - 2 = 4\left[\ln(x+1) - \frac{1}{2}\right]$$

(B1)

Translation

(M1)

$$\begin{bmatrix} -1 \\ -\frac{1}{2} \end{bmatrix}$$

(A1)

Stretch I

I + (II or III)

(M1)

SF 4 II

// y-axis III

I + II + III

(A1)

All correct and no mistakes on order etc

(A1)

(6)

(ii) $y = 4\ln(x + 1) - 2$

$x = 0$ $y = -2$

B1

$y = 0$
$4\ \ln(x + 1) = 2$
$\ln(x + 1) = \frac{1}{2}$

isolate $\ln(x + 1) =$ or $(x + 1)^4$

M1

$x + 1 = e^{\frac{1}{2}}$

$x + 1 = e^k$

A1

$x = e^{\frac{1}{2}} - 1$ o.e.

CSO isw

A1

4

[12]

Q29.

(a) $3\log_a x = \log_a 8 \Rightarrow \log_a x^3 = \log_a 8$

OE use of the log law

M1

$x^3 = 8 \Rightarrow x = 2$

A1

2

(b) $3\ \log_a 6 - \log_a 8 = \log_a 6^3 - \log_a 8$

Correct use of one log law

M1

$= \log_a \frac{6^3}{8}$

Correct use of a different log law

M1

$= \log_a \frac{216}{8} = \log_a 27$

*CSO **AG** (be convinced)*

A1

3

(c) (i) {p = } $3\log_{10} 3 - \log_{10} 8$

Substitute x = 3

M1

$$p = \log_{10}\frac{3^3}{8} = \log_{10}\frac{27}{8}$$

***AG** (be convinced)*

A1

2

(ii) Gradient of PQ = $\frac{q-p}{6-3}$

used $\frac{\text{difference in } y\text{-coords}}{\text{difference in } x\text{-coords}}$

M1

$$\ldots\ldots\ldots. = \frac{\log_{10} 27 - \log_{10}\frac{27}{8}}{3}$$

Any correct exact form

A1

$$\ldots\ldots\ldots = \frac{1}{3}\log_{10}\left(27 \div \frac{27}{8}\right)$$

Correct use of log law

m1

Gradient = $= \frac{1}{3}\log_{10} 8 = \log_{10} 2$

***AG** (be convinced)*

A1

4

[11]

Q30.

(a) $y_A = 3\,(2^0 + 1)$

Substituting x = 0 PI

M1

= 6

A1

2

(b) $h = 2$

PI

B1

Integral = h /2 {……}
{….} = f(0) + 2[f(2) + f(4)] + f(6)

OE summing of areas of the three traps.

M1

{} = 6 + 2[3 × 5 + 3 × 17] + 3 × 65 = 6 + 2[15 + 51] + 195

Condone 1 numerical slip {ft on (a) for f(0) if not recovered}

[Sum of 3 traps. = 21 + 66 + 246]

A1

Integral = 333

CAO

A1

4

(c) (i) $21 = 3(2^x + 1) \Rightarrow 2^x = 6$

AG (be convinced)

B1

1

(ii) $\log_{10}2^x = \log_{10} 6$

Take ln or $\log_{10}$ of both sides of $a^x = b$ or other relevant base if clear. The equation $a^x = b$ used must be correct.

M1

$x\log_{10}2 = \log_{10}6$

Use of log $2^x = x$ log 2 OE

m1

$x = \dfrac{\lg 6}{\lg 2}$ = 2.5849....= 2.58 to 3sf

Both method marks must have been awarded.

A1

3

[10]

Q31.

(a) $\log_a n = \log_a 3(2n - 1)$

OE Log law used PI by next line

M1

$\Rightarrow n = 3(2n - 1)$

*OE, but must **not** have any logs.*

m1

$\Rightarrow 3 = 5n \Rightarrow n = \frac{3}{5}$

A1

3

(b) (i) $\log_a x = 3 \Rightarrow x = a^3$

B1

1

(ii) $\log_a y - \log_a 2^3 = 4$

3 log 2 = log 2^3 seen or used any time in (ii)

M1

$$\log_a \frac{y}{2^3} = 4 \left\{ \begin{array}{l} xy = a^7 \times a^{(3\log_a 2)} \\ \text{or} \\ y = a^4 \times a^{(3\log_a 2)} \end{array} \right\}$$

*Correct method leading to an equation involving y (or xy) and a log but **not** involving + or −*

M1

$$\frac{y}{2^3} = a^4 \quad \left\{ \begin{array}{l} xy = a^7 \times 2^3 \\ \text{or} \\ y = a^4 \times 2^3 \end{array} \right\}$$

*Correct method to eliminate **ALL** logs e.g. using $\log_a N = k \Rightarrow N = a^k$ or using $a^{\log_a c} = c$*

ml

$by = a^3 \times 8a^4$ or $8a^7$

A1

4

[8]

Q32.

(a) (i) $t = 0$: $x = 3$

B1 **1**

(ii) $t = 14$: $x = 15 - 12e^{-1}$

or $15 - 12e^{\frac{-14}{14}}$

M1

$= 10.6$

CAO

A1 **2**

(b) (i) $-5 = -12e^{-\frac{t}{14}}$

substitute $x = 10$*; rearrange to form* $p = qe^{-\frac{t}{14}}$

M1

$\ln\left(\frac{5}{12}\right) = -\frac{t}{14}$ (OE)

take lns correctly

m1

$t = 14\ln\left(\frac{12}{5}\right)$

must come from correct working

A1 **3**

(ii) $t = 12.256... \approx 12$ days

ft on a, b *if* $a > b$*; accept t = 12 NMS*
Accept 12 from incorrect working in (b)(i)
Accept 13 if 12.2 or 12.3 seen

B1F **1**

(c) (i) $\frac{dx}{dt} = -\frac{1}{14}x - 12e^{-\frac{t}{14}}$

differentiate; allow sign error

condone $\frac{dy}{dx}$ *used consistently*

M1

$= -\frac{1}{14}(x - 15)$

Or $\frac{1}{14}\left(12e^{-\frac{t}{14}}\right)$ *and* $12e^{-\frac{t}{14}}$ *= 15 − x seen*

m1

$= \frac{1}{14}(15 - x)$

AG − be convinced CSO

A1 3

Alt: $t = -14\ln\left(\frac{15-x}{12}\right)$

attempt to solve given equation for t

(M1)

$\frac{dt}{dx} = \frac{-14\left(-\frac{1}{12}\right)}{\left(\frac{15-x}{12}\right)}$

differentiate wrt x, with $\frac{1}{\frac{15-x}{12}}$ *seen; OE*

(m1)

$\frac{dt}{dx} = \frac{14}{15-x} \Rightarrow \frac{dx}{dt} = \frac{1}{14}(15-x)$

AG − be convinced

(A1) (3)

Alt: (backwards)

$$\int \frac{dx}{15-x} = \int \frac{dt}{14} = \pm 14\ln(15-x) = t + c$$

(M1)

Use (0,3): $-14 \ln (15 - x) + 14 \ln 12 = t$

(m1)

Solve for x: $x = 15 - 12e^{-\frac{t}{14}}$

All steps shown

(A1)

(3)

(ii) rate of growth = 0.5 (cm per day)

Accept $\frac{7}{14}$

B1

1

[11]

Q33.

(a) $2\log_a n - \log_a (5n - 24) = \log_a 4$

$\Rightarrow \log_a n^2 - \log_a (5n - 24) = \log_a 4$

A law of logs used

M1

$\Rightarrow \log_a \left[\frac{n^2}{5n-24}\right] = \log_a 4$

A second law of logs used leading to both sides being single log terms or single log term on LHS with RHS = 0

M1

$\Rightarrow \frac{n^2}{5n-24} = 4$

$\Rightarrow n^2 - 20n + 96 = 0$

CSO. *AG*

A1

3

(b) $\Rightarrow$ $(n - 8)(n - 12) = 0$

Accept alternatives eg formula, completing of sq..

M1

$\Rightarrow$ $n = 8,\ 12$

A1

2

[5]

Q34.

(a) $a = -8$

B1

$e^{2x} - 9 = 0$

M1

$e^{2x} = 9$

$2x = \ln 9$

$x = \ln 3$

AG Condone verfication

A1

3

(b) $\left(e^{2x} - 9\right)^2 = e^{4x} - 18e^{2x} + 81$

AG

B1

1

[4]

Q35.

(a) $y = x^{-2} \ln x$

$\frac{dy}{dx} = x^{-2}\,\frac{1}{x} - 2x^{-3}$ $\ln x$

Use of product or quotient

M1

each term

A1 A1

$= \dfrac{1 - 2\ln x}{x^3}$

Convincing argument $x^{-2} \times \dfrac{1}{x} = x^{-3}$
AG

A1 **4**

(b) $\int x^{-2} \ln x \, dx$ $\quad u = \ln x \quad dv = x^{-2}$
Attempt at integration by parts

M1

$du = \dfrac{1}{x} \qquad v = -x^{-1}$

A1

$\int = -\dfrac{1}{x}\ln x + \int x^{-2}\, dx$

A1

$= -\dfrac{1}{x}\ln x - \dfrac{1}{x} (+c)$

A1 **4**

(c) (i) At A, $\dfrac{dy}{dx} = 0$

$1 - 2\ln x = 0$

$\ln x = \dfrac{1}{2}$
Attempt at ln $x = k$

M1

$x = e^{\frac{1}{2}}$

A1 **2**

(ii) $R = \left[-\dfrac{1}{x}(\ln x + 1)\right]_1^5$

$R = \left[\text{Their (b)}\right]_1^5$

M1

$= -\frac{1}{5}(\ln 5 + 1) + (\ln 1 + 1)$

OE

A1

$= \frac{1}{5}(4 - \ln 5)$

convincing argument; AG

A1

3

[13]